Efficacy and Safety of HRS9531, a Novel Dual GLP-1/GIP Receptor Agonist, in Chinese Adults with Obesity without Diabetes—Up to 52-Week Treatment

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85TH SCIENTIFIC SESSIONS

JUNE 20–23, 2025
CHICAGO, IL

Abstract: #2566

American Diabetes Association

Financial Disclosure

- The principal investigator, Professor Xiaoying Li, reports no relevant financial relationships to disclose.
- This study is sponsored by Jiangsu Hengrui Pharmaceuticals Co., Ltd (China).

BACKGROUND/OBJECTIVE

- Dual GLP-1/GIP receptor agonists (RAs) are emerging as a key therapeutic option in obesity treatment [1].
- HRS9531, a novel dual GLP-1/GIP RA, previously demonstrated clinically significant weight loss in obese adults over 24 weeks [2] and is currently in Phase 2/3 development for obesity, type 2 diabetes mellitus (T2DM), and obstructive sleep apnea.
- This study assessed the long-term efficacy and safety of HRS9531 in Chinese adults with obesity without diabetes over a treatment period of up to 52 weeks.

References:

[1] Neumiller JJ, et al. Diabetes Care. 2025 Feb 1;48(2):177-181.

[2] Lin Zhao, et al. Diabetes 2024;73(Supplement_1):1861-LB.

Methods and Materials

- Chinese adults (BMI 28–40 kg/m²) were randomized 1:1:1:1:1 to receive once-weekly (QW) subcutaneous HRS9531 (1.0, 3.0, 4.5, or 6.0 mg) or placebo during a 24-week core treatment period, followed by a 28-week extension (ClinicalTrials.gov identifier: NCT05881837). Results from the core period were previously reported [2].
- During the 28-week extension period, participants maintained their original treatments and doses for the first 8 weeks. Subsequently, all received HRS9531 for 20 weeks: those initially assigned to placebo or the 1.0 mg group transitioned to a 3.0 mg QW dose, while participants in other dose groups were re-randomized to receive either QW or biweekly (every two weeks; Q2W) dosing **(Figure 1).**

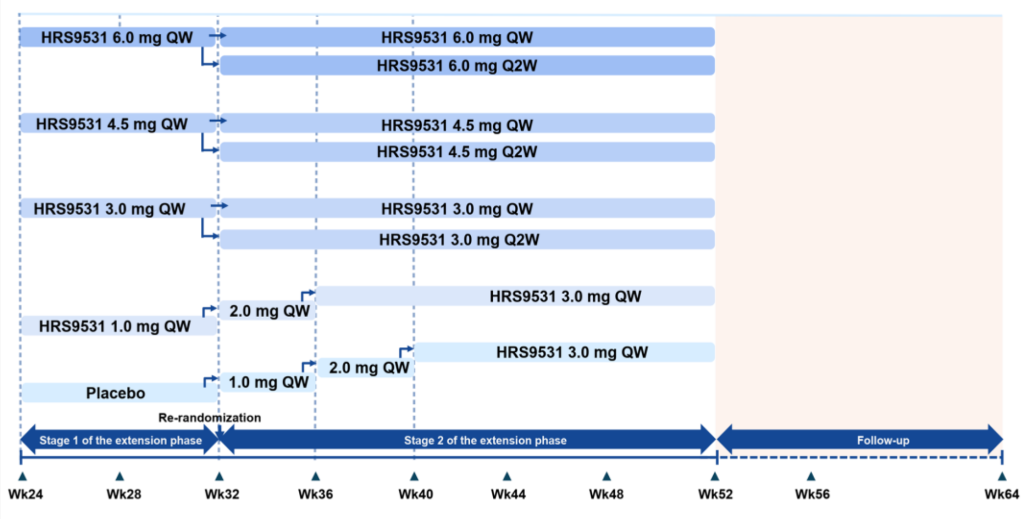

Figure 1. Study design of the extension phase

Results

Participants

- Of 249 enrolled participants, 248 received at least one post-randomization dose of the study treatment, and 240 completed the 24-week core period.
- Subsequently, 168 participants entered the optional extension phase, of whom 166 completed the first 8 weeks of their original treatment. Thereafter, 163 participants were re-randomized to receive HRS9531 for a 20-week extension, and 154 completed this treatment.
- During the extension phase (both Stage 1 and Stage 2), no participants discontinued treatment because of an adverse event.

Efficacy

- At week 32, across HRS9531 doses, the mean percentage change in body weight from baseline ranged up to -18.0% in the 6.0 mg weekly (QW) group, compared to a 0.7% increase in the placebo group **(Figure 2)**.
- Proportions achieving ≥5% and ≥10% body weight reductions reached as high as 100% and 88.5% in HRS9531 dose groups, respectively, versus 12.5% and 9.4% in the placebo group **(Figure 3)**.

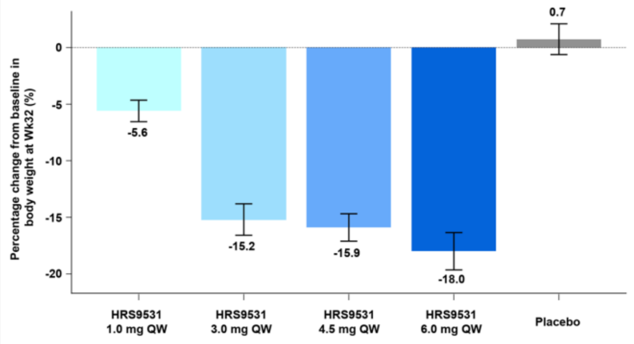

Figure 2. Percentage change in body weight from baseline at Wk32 (Mean [SE]; observed cases)

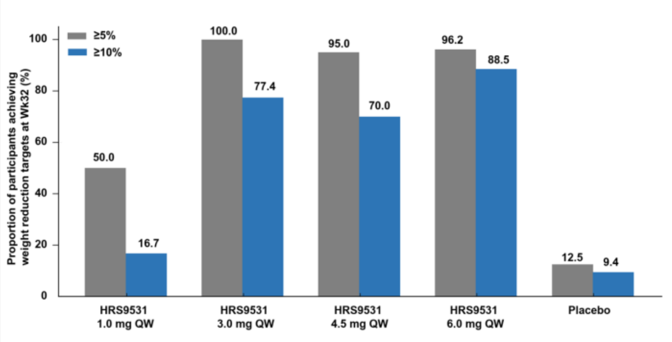

Figure 3. Proportion of participants achieving weight reduction targets at Wk32 (observed cases)

Results

Efficacy

- For waist circumference, the mean reduction from baseline reached up to - 13.8 cm in the 6.0 mg QW group, versus -2.0 cm in the placebo group **(Figure 4).**
- From week 32 to week 52, the mean percentage change in body weight ranged from -0.76% to 0.01% in the HRS9531 Q2W groups versus -3.45% to -0.02% in the HRS9531 QW groups **(Figure 5)**.
- At week 32, HRS9531 outperformed placebo in improving BMI, blood pressure, and glycemic control, as well as in reducing triglyceride, uric acid, and ALT levels **(Table 1).**
- The Q2W regimen maintained weight stability, suggesting that less frequent dosing may be adequate for long-term weight loss maintenance.

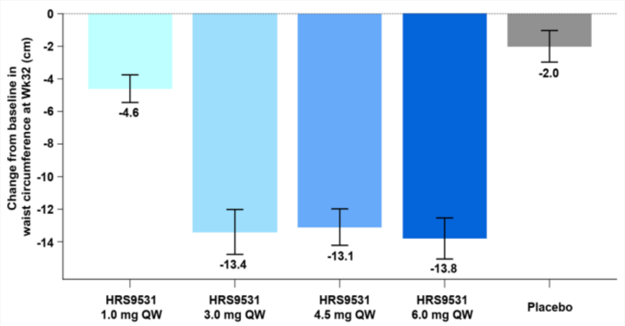

Figure 4. Changes in waist circumference from baseline at Wk32 (Mean [SE]; observed cases)

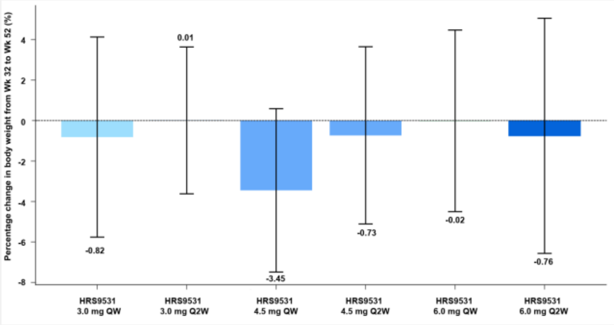

Figure 5. Mean (SE) percentage change in body weight from Wk32 during the 20-week extension phase (Wk52; observed cases)

"HRS9531 3.0 mg QW" refers only to participants who received HRS9531 3.0 mg QW during the core period and stayed on that regimen after re-randomization.

Table 1. Change from baseline at Wk32

	HRS9531 1.0 mg QW (n=37)	HRS9531 3.0 mg QW (n=32)	HRS9531 4.5 mg QW (n=40)	HRS9531 6.0 mg QW (n=26)	Placebo (n=33)
BMI, kg/m^2*	-1.8 (1.9)	-5.1 (2.6)	-5.2 (2.6)	-5.8 (2.8)	0.3 (2.4)
SBP, mmHg*	-3.6 (9.5)	-7.5 (13.2)	-9.0 (9.0)	-10.8 (9.9)	0.3 (7.3)
DBP, mmHg*	-3.5 (7.4)	-3.2 (8.6)	-4.1 (7.9)	-6.8 (7.8)	-0.5 (6.9)
HbA_{1c}, %*	-0.2 (0.2)	-0.4 (0.2)	-0.4 (0.2)	-0.4 (0.2)	0.1 (0.2)
TG, %#	-15.9 (32.9)	-28.9 (22.3)	-28.1 (30.6)	-33.4 (24.5)	2.2 (34.9)
Uric acid, %#	-16.6 (13.7)	-17.5 (20.8)	-19.8 (15.8)	-18.9 (14.5)	1.0 (17.9)
ALT, %#	-22.0 (36.8)	-31.4 (45.4)	-17.3 (88.0)	-30.8 (36.2)	27.6 (96.4)

"n" refers to the number of participants who received treatment in Stage 1 of the extension phase.
*Data are value changes from baseline at W32 and presented in observed mean (SD).
#Data are percentage changes from baseline at W32 and presented in observed mean (SD).
SBP, systolic blood pressure; DBP, diastolic blood pressure; TG, triglycerides; ALT, alanine aminotransferase.

Results

Safety

- During the 32-week placebo-controlled treatment period, treatment-emergent adverse events (TEAEs) were reported in 75.5% to 91.8% of participants across HRS9531 dose groups, compared with 85.7% in the placebo group **(Table 2)**.
- Most TEAEs were mild to moderate in severity, with gastrointestinal events being the most common.
- No new safety signals emerged for HRS9531 through 52 weeks of treatment, and its tolerability mirrored that of established GLP-1/GIP RAs.

Table 2. Treatment-emergent adverse events during the 32-week placebo-controlled treatment period

	HRS9531 1.0 mg QW (n=49)	HRS9531 3.0 mg QW (n=51)	HRS9531 4.5 mg QW (n=50)	HRS9531 6.0 mg QW (n=49)	Placebo (n=49)
Any TEAEs	37 (75.5)	45 (88.2)	40 (80.0)	45 (91.8)	42 (85.7)
TESAEs	1 (2.0)	2 (3.9)	3 (6.0)	0	3 (6.1)
TEAEs leading to treatment discontinuation	1 (2.0)	1 (2.0)	0	0	1 (2.0)
Gastrointestinal disorders with ≥5% frequency in any arm					
Nausea	8 (16.3)	14 (27.5)	16 (32.0)	16 (32.7)	4 (8.2)
Diarrhea	5 (10.2)	18 (35.3)	15 (30.0)	16 (32.7)	4 (8.2)
Vomiting	3 (6.1)	10 (19.6)	12 (24.0)	14 (28.6)	1 (2.0)
Abdominal distension	1 (2.0)	9 (17.6)	3 (6.0)	4 (8.2)	0
Dyspepsia	0	4 (7.8)	1 (2.0)	1 (2.0)	0
Constipation	1 (2.0)	3 (5.9)	1 (2.0)	3 (6.1)	0
Abdominal pain	0	3 (5.9)	4 (8.0)	1 (2.0)	0
Eructation	0	2 (3.9)	2 (4.0)	4 (8.2)	0

Data are n (%).
TEAE, treatment-emergent adverse event; TESAEs, treatment-emergent serious adverse events.

CONCLUSION

- In Chinese adults with obesity without diabetes, HRS9531 demonstrated clinically significant body weight reduction compared to placebo over the 32-week treatment period.
- The Q2W (every 2 weeks) dosing regimen effectively sustained the weight loss achieved during the initial 32-week phase through week 52.
- HRS9531 exhibited a tolerable and manageable safety profile, aligning with the established safety patterns of other GLP-1/GIP RAs.